

OSI Layer and Elastic Load Balancer

Class no:19

What is a Load Balancer?

A **load balancer** is a networking device or service that acts as an intermediary between clients and servers. Its primary function is to **distribute incoming network traffic across multiple servers** to ensure no single server is overwhelmed, improving:

- **Availability:** Keeps services running even if some servers fail.
- **Scalability:** Handles increased traffic by routing requests to multiple targets.
- **Performance:** Reduces response times by evenly distributing workloads.
- **Reliability:** Monitors server health and routes traffic only to healthy targets.

How it works:

1. Receives traffic on defined **ports and protocols** (HTTP, HTTPS, TCP, UDP).
2. Selects a server from a **target group** using algorithms like round-robin, least connections, or source IP hash.
3. Forwards requests to the target and sends the processed response back to the client.
4. Continuously performs **health checks** on servers to ensure traffic only reaches available targets.

OSI Model (Open Systems Interconnection)

The OSI model is a **conceptual framework for networked systems**, dividing telecommunications functions into seven layers:

1. **Physical Layer (L1):** Transmission of raw binary data over a medium.
2. **Data Link Layer (L2):** Error detection, MAC addressing, frame delivery.
3. **Network Layer (L3):** Routing and IP addressing.
4. **Transport Layer (L4):** Reliable data transfer, TCP/UDP communication, session multiplexing.
5. **Session Layer (L5):** Managing and maintaining application sessions.
6. **Presentation Layer (L6):** Data translation, encryption/decryption, compression.
7. **Application Layer (L7):** Interface for end-user applications; HTTP, HTTPS, DNS, etc.

Load Balancing across OSI Layers:

- **Layer 3 (Network Layer):** Routes traffic based on IP addresses. Minimal overhead, high throughput.
- **Layer 4 (Transport Layer):** Uses TCP/UDP headers for routing. Efficient for session management; limited application awareness.
- **Layer 7 (Application Layer):** Routes based on content (HTTP headers, URLs, cookies). Intelligent, granular routing, supports advanced features like SSL termination, WebSockets, and content-based routing.

Types of Load Balancers in AWS and Their OSI Layer Differences

AWS provides **four primary load balancers**:

Type	OSI Layer	Protocols
Application Load Balancer (ALB)	7 (Application Layer)	HTTP, HTTPS, WebSocket, gRPC
Network Load Balancer (NLB)	4 (Transport Layer)	TCP, UDP, TLS
Gateway Load Balancer (GWLB)	3 & (appliance-aware)	IP Packets (transparent forwarding)
Classic Load Balancer (CLB)	4 & 7	TCP, HTTP, HTTPS

Key Differences:

1. **OSI Layers:** ALB (L7) > intelligent content-based routing; NLB (L4) > connection-based, fast; GWLB (L3) > virtual appliance scaling; CLB (L4/L7) > legacy, fewer features.
2. **Routing Logic:** ALB routes by URL, headers, or queries; NLB routes by IP/port; GWLB forwards packets to appliances; CLB uses basic load distribution.
3. **Features & Use Cases:** ALB is best for web apps with complex routing; NLB for low-latency high-throughput traffic; GWLB for security appliances; CLB for simpler or legacy workloads.

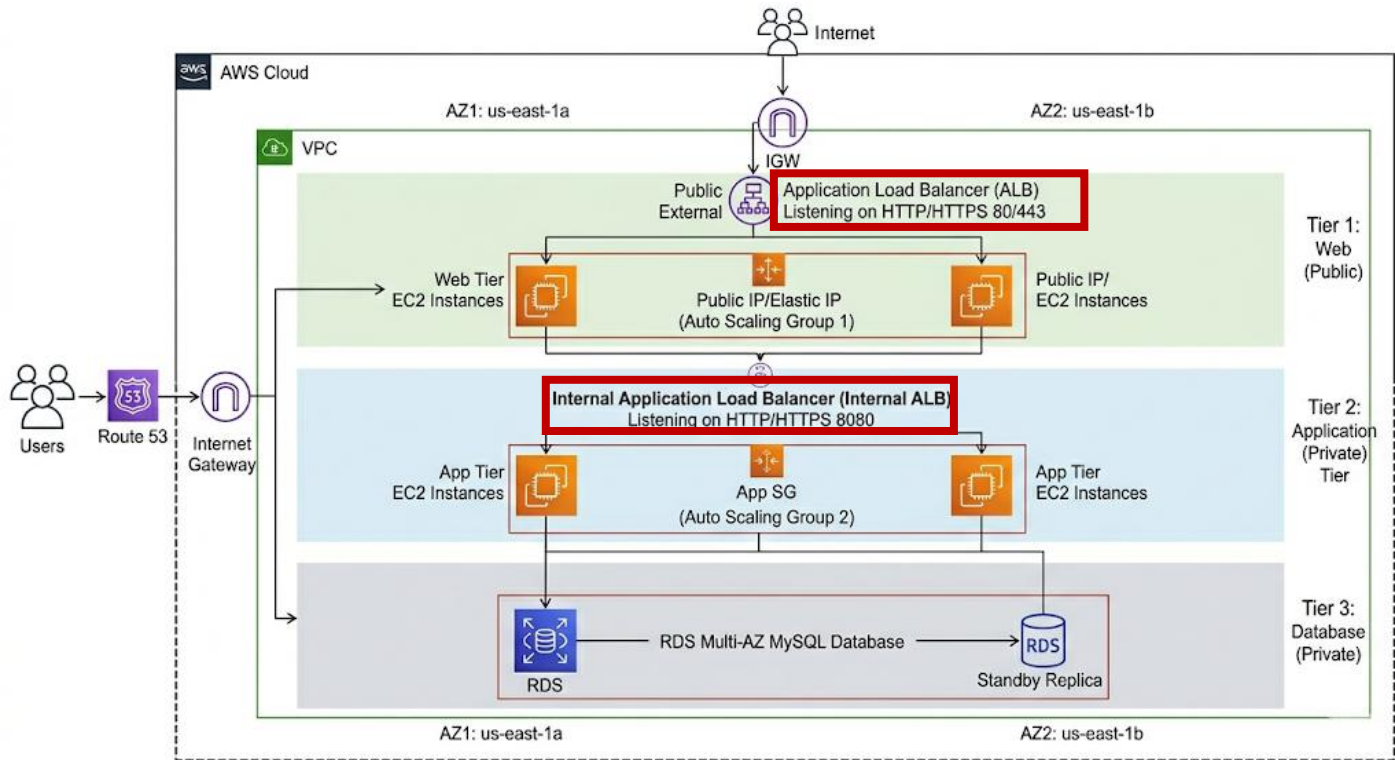


Fig. Architecture of a three-tier application with EC2 and Load balancers

Hands-on: Creating a load balancer with two Ec2 instances

Step no:1 Create & Launch Two EC2 Instances

- Name: my-LB-instance1 & my-LB-instance2
- Use Amazon Linux 2 AMI.
- Security group → select existing group → default.

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0538c231a226fe3c9 (default) [↻](#)

Subnet [Info](#)

No preference [↻](#) [Create new subnet](#)

Availability Zone [Info](#)

No preference [↻](#) [Enable additional zones](#)

Auto-assign public IP [Info](#)

Enable [↻](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups [↻](#) [Compare security group rules](#)

default sg-06d80e515c32b9515 [✕](#)

VPC: vpc-0538c231a226fe3c9

Security groups that you add or remove here will be added to or removed from all your network interfaces.

- In the **Advanced Details** → **User Data**,
paste your script → launch two instances.

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
# Update package repository and install Apache
sudo yum update -y
sudo yum install httpd -y

# Start the Apache and enable it to start on boot
sudo systemctl start httpd
sudo systemctl enable httpd

# Create an index.html file with the hostname
echo "<html><head><title>welcome to $(hostname)</title></head></body>
<h1> hello from $(hostname)</h1></body></html>" | sudo tee /var/www/html/index.html

# Restart apache to apply changes
sudo systemctl restart httpd
```

☐ User data has already been base64 encoded

Step no:02 Open Security Group Ports

Ensure both EC2 instances' security groups allow:

- **Inbound HTTP (port 80)** from anywhere (0.0.0.0/0).
- **Inbound SSH (port 22)** for admin access.
- **All TCP (0-65535)** It will enable (SSH as well as HTTP and HTTPS. They all are part of the TCP.)

[Security Groups](#) > [sg-06d80e515c32b9515 - default](#) > [Edit inbound rules](#)

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info					
Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
sg-r-06b9fe308e8427a99	All traffic	All	All	Custom	Q Delete
sg-r-02321ea3f8cbc9385	SSH	TCP	22	Custom	sg-06d80e515c32b9515 Delete
sg-r-0db828bb575c27c13	HTTPS	TCP	443	Custom	0.0.0.0 Delete
sg-r-0551be0e5d59eb682	HTTP	TCP	80	Custom	0.0.0.0 Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

sg-06d80e515c32b9515 - default

Details

Security group name
default

Security group ID
sg-06d80e515c32b9515

Description
default VPC security group

VPC ID
[vpc-0538c231a226fe3c9](#)

Owner
314383175192

Inbound rules count
5 Permission entries

Outbound rules count
1 Permission entry

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (5)

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0455473733711f1b3	IPv4	SSH	TCP	22	0.0.0.0/0	-
☐	-	sgr-06b9fe308e8427a99	-	All traffic	All	All	sg-06d80e515c32b951...	-
☐	-	sgr-0fad7043be9d4af77	IPv4	HTTPS	TCP	443	0.0.0.0/0	-
☐	-	sgr-0cc66589f73127520	IPv4	HTTP	TCP	80	0.0.0.0/0	-
☐	-	sgr-0dc082f6cc5b380fb	IPv4	All TCP	TCP	0 - 65535	0.0.0.0/0	-

Step no:03 Verify Apache Installation in our instances by,

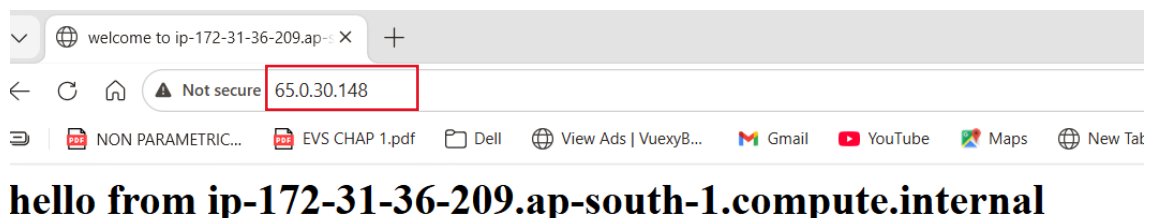
- After launch, connect via browser: `http://<Public_IP_of_Instance>` in a private window.

Public ip of my-LB-instance1:



You should see: *hello from <hostname>*. (hostname is DNS+some ip address)

Public ip of my-LB-instance2:



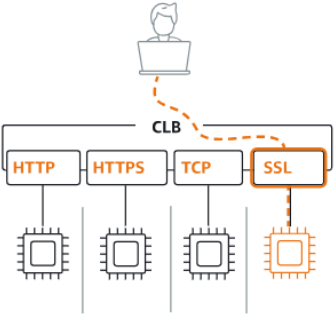
You should see: *hello from <hostname>*. (hostname is DNS+some ip address)

Step no:4 Create Classic Load Balancer

- Go to EC2 → Load Balancers → Create Load Balancer → Classic Load Balancer.

▼ Classic Load Balancer - *previous generation*

Classic Load Balancer [Info](#)



Classic Load Balancers are the previous generation of load balancers from you migrate to a current generation load balancer. For more information, [Load Balancer](#).

[Create](#)

- Provide a name (e.g., Myclb). → choose the subnets.

Create Classic Load Balancer [Info](#)

The Classic Load Balancer distributes incoming application traffic across multiple EC2 Instance targets in multiple Availability Zones. This increases the fault tolerance of your applications. Elastic Load Balancing detects unhealthy instances and routes traffic only to healthy instances.

► How Classic Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

Myclb

Maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your network settings.

VPC [Info](#)

Select the virtual private cloud (VPC) for your targets. Only VPCs with an internet gateway are available for selection. The selected VPC cannot be changed after the load balancer is created. When selecting a VPC for your load balancer, ensure each subnet has a CIDR block with at least a /27 bitmask and at least 8 free IP addresses.

[Learn more](#)

vpc-0538c231a226fe3c9
172.31.0.0/16

(default)



[Create VPC](#)

Availability Zones and subnets

Select at least one Availability Zone and one subnet for each zone. We recommend selecting at least two Availability Zones. The load balancer will route traffic only to targets in the selected Availability Zones. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-01bf4d440c0912db3
172.31.32.0/20

IPv4 address

Assigned by AWS

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0b3c21b6de4314d95
172.31.0.0/20

IPv4 address

Assigned by AWS

☒ ap-south-1c (aps1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c305b80f72541592
172.31.16.0/20

IPv4 address

Assigned by AWS

- Select **HTTP (port 80)** as the listener.
- Assign a security group that allows inbound HTTP.

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

default (sg-06d80e51sc32b515) X

Listeners and routing Info

A listener is a process that checks for connection requests using the protocol and port you configure. The settings you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Instance HTTP:80

Remove

Listener protocol

HTTP

Instance protocol

HTTP

Listener port

80

1-65535

Instance port

80

1-65535

Add listener

You can add up to 99 more listeners.

Health checks Info

Your load balancer automatically performs health checks to test the availability of all registered instances. Traffic is only routed to healthy instances, which is determined on their response to the health check.

Ping target

The health check ping is sent using the protocol and port you specify. If using HTTP/HTTPS protocol, you must also provide the destination path.

Ping protocol

HTTP

Ping path

/index.html

Ping port

80

1-65535

► Advanced health check settings

- Instance, add instance → select our instances → my-LB-instance1 & my-LB-instance2

Instances (0)

You can add instances to register as targets of the load balancer. Alternatively, after your load balancer is created, you can add it to an Amazon EC2 Auto Scaling group to ensure you maintain the correct number of instances to handle the load for your application. For maximum fault tolerance, we recommend maintaining approximately equivalent numbers of instances in each Availability Zone.

Filter instances

Instance ID Name State Security groups Zone Public IPv4 address Subnet ID

No instances added

Remove

Add instances

Add instances

Select EC2 instances to register to your load balancer. Requests will be routed to registered instances that meet the health check requirements. For maximum fault tolerance, we recommend maintaining approximately equivalent numbers of instances in each Availability Zone enabled for the load balancer. If demand on your instances changes, you can register or deregister instances without disrupting the flow of requests to your application. [Learn more](#)

VPC
vpc-0538c231a229f3d9

Available instances (2/2)

Filter available instances

Instance ID	Name	State	Security groups	Zone	Public IPv4 address	Subnet ID	Launch time
i-0891b651ec27ca50	my-LB-instance1	Running	default	ap-south-1a	43.205.135.181	subnet-01f64440c0912db3	April 6, 2026, 20:03 (UTC+05:30)
i-5502d972b0602afd	my-LB-instance2	Running	default	ap-south-1a	65.0.30.148	subnet-01f64440c0912db3	April 6, 2026, 20:03 (UTC+05:30)

Cancel

Confirm

- Create Load Balancer.

Instances (2/2)

You can add instances to register as targets of the load balancer. Alternatively, after your load balancer is created, you can add it to an Amazon EC2 Auto Scaling group to ensure you maintain the correct number of instances to handle the load for your application. For maximum fault tolerance, we recommend maintaining approximately equivalent numbers of instances in each Availability Zone.

Filter instances

☒

Instance ID

☒

I-0891bb51eec27ca50

☒

I-0502d5972b0602afd

my-Lb-instance1

my-Lb-instance2

Running

Running

default

default

ap-south-1a

ap-south-1a

43.205.135.181

65.0.30.148

subnet-01bf4

subnet-01bf4

Attributes

Creating your load balancer using the console gives you the opportunity specify additional features at launch. You can also find and adjust these settings in the load balancer's "Attributes" section after your load balancer is created.

☒

Enable cross-zone load balancing

With cross-zone load balancing, each load balancer node for your Classic Load Balancer distributes requests evenly across the registered instances in all enabled Availability Zones. If cross-zone load balancing is disabled, each load balancer node distributes requests evenly across the registered instances in its Availability Zone only. Classic Load Balancers created with the API or CLI have cross-zone load balancing disabled by default. After you create a Classic Load Balancer, you can enable or disable cross-zone load balancing at any time.

☒

Enable connection draining

Applicable to instances that are deregistering, this feature allows existing connections to complete (during a specified draining interval) before reporting the instance as deregistered. [Learn more](#)

Timeout (draining interval)

The maximum time for the load balancer to allow existing connections to complete. When the maximum time limit is reached, the load balancer forcibly closes any remaining connections and reports the instance as deregistered.

300

seconds

Valid values: 1-3600 (integers only)

Load balancer tags - optional

Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them. The 'Key' is required, but 'Value' is optional. For example, you can have Key = production-webserver, or Key = webserver, and Value = production.

Review

Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose **Create load balancer**.

Summary

Review and confirm your configurations. [Estimate cost](#)

Basic configuration [Edit](#)

Name: Myclb

Scheme: Internet-facing

Network mapping [Edit](#)

VPC: [vpc-0538c231a226fe3c9](#)

Availability Zones and subnets:

- ap-south-1a
[subnet-01bf4d440c0912db3](#)
- ap-south-1b
[subnet-0b3c21b6de4314d95](#)
- ap-south-1c
[subnet-0c305b80f72541592](#)

Security groups [Edit](#)

default

[sg-06d80e515c32b9515](#)

Listeners and routing [Edit](#)

HTTP:80

Health checks [Edit](#)

HTTP:80/index.html

- Timeout: 2 seconds
- Interval: 5 seconds
- Unhealthy threshold: 2
- Healthy threshold: 10

Instances [Edit](#)

2 instances added

- 2 instances in ap-south-1a

Attributes [Edit](#)

- Cross-zone load balancing: On
- Connection draining: On
- Connection draining timeout: 300 seconds

Tags [Edit](#)

-

Cancel

Create load balancer

Step no:05 Once the load balancer has been created, wait until the status of the instances in it become active.

Then copy the DNS paste it in a private tab.

Myclb

▼ Details

Load balancer type

Classic

Scheme

Internet-facing

Status

2 of 2 instances in service

Hosted zone

ZP97RAFLXTNZK

VPC

[vpc-0538c231a226fe3c9](#)

Date created

April 6, 2026, 20:24 (UTC+05:30)

Availability Zones

[subnet-01bf4d440c0912db3](#) ap-south-1a (aps1-az1)
[subnet-0c305b80f72541592](#) ap-south-1c (aps1-az2)
[subnet-0b3c21b6de4314d95](#) ap-south-1b (aps1-az3)

DNS name [Info](#)

[Myclb-1637517480.ap-south-1.elb.amazonaws.com](#) (A Record)

This Classic Load Balancer can be migrated to a next generation load balancer. Migration wizard uses your load balancer's current configurations to create a new load balancer. [Learn more](#)

Launch migration wizard

Distribution of targets by Availability Zone (AZ)

For each enabled Availability Zone, you can view the number of registered instances and their current health states. Selecting any values here will apply the corresponding filter to the Target instances table.

We can see the output there, refresh the page we can see the traffic is now passed to the next instance. It shows load balancer is distributing the traffic across the two instances we have created.

welcome to ip-172-31-47-173.ap-south-1.elb.amazonaws.com

Not secure myclb-1637517480.ap-south-1.elb.amazonaws.com

NON PARAMETRIC... EVS CHAP 1.pdf Dell View Ads | VuexyB... Gmail YouTube Maps New Tab

hello from ip-172-31-47-173.ap-south-1.compute.internal

welcome to ip-172-31-36-209.ap-south-1.elb.amazonaws.com

Not secure myclb-1637517480.ap-south-1.elb.amazonaws.com

NON PARAMETRIC... EVS CHAP 1.pdf Dell View Ads | VuexyB... Gmail YouTube Maps New Tab

hello from ip-172-31-36-209.ap-south-1.compute.internal